Notes on "Topology and Geometry" by Bredon

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1 Metric Spaces

1.1 Definition of Metric Space

A metric space is a set X with a function $d : X \times X \to \mathbb{R}$ called the metric that satisfies the following properties

1. $d(x, y) \geq 0$ for all $x, y \in X$
2. $d(x, y) = d(y, x)$ for all $x, y \in X$
3. $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$

The ϵ ball about a point $x \in X$ is defined to

$$B_\epsilon(x) = \{p \in X, d(x, p) < \epsilon\}$$

1.2 Proposition

A function $f : X \to Y$ between metric spaces is continuous if and only if $f^{-1}(U)$ is open in X for each open subset U of Y.

Proof: Suppose that f is continuous and U is an open subset of Y. Let $f(x) \in U$. Since U is an open set, then there exists an open ball $B_\epsilon(f(x))$ that is completely contained in U. Recall that the a function is continuous if for an arbitrary $\epsilon > 0$, there exists a $\delta > 0$ such that $d_X(x_1, x_2) < \delta$ implies $d_Y(f(x_1), f(x_2)) < \epsilon$. Now consider a point $y_1 = f(x_1) \in B_\epsilon(f(x))$. Clearly, $d_Y(y, y_1) = d_Y(f(x), f(x_1)) < \epsilon$ and so $d_X(x_1, x_2) < \delta$, and so there exists a δ ball about $x \in X$, $B_\delta(x)$, and clearly $B_\delta(x) \subset f^{-1}(U)$, so $f^{-1}(U)$ is open in X.

To show the converse, suppose that $f^{-1}(U)$ is open in X for each open subset U of Y. Consider any point $y = f(x)$ and an $epsilon > 0$. The corresponding ϵ ball is $B_\epsilon(y)$. From the assumption, we know there is a corresponding open set $f^{-1}(B_\epsilon(y))$. Furthermore, we know that $x \in f^{-1}(B_\epsilon(y))$ and so we can deduce that there exists an open δ ball contained entirely within this set, i.e. $B_\delta(x) \subset f^{-1}(B_\epsilon(y))$. We thus conclude that for the $\epsilon > 0$, there exists a $\delta > 0$ such that for any point $p \in B_\delta(x)$, $f(p) \in B_\epsilon(y)$, which is the definition of continuity of f.

1.3 Proposition

If d_1 and d_2 are metrics on the same set X which satisfy the condition that for any two points $x, y \in X$ and $\epsilon > 0$, there is a $\delta > 0$ such that

$$d_1(x,y) < \delta \implies d_2(x,y) < \epsilon$$
$$d_2(x,y) < \delta \implies d_1(x,y) < \epsilon$$

then the metrics define the same open set on X.

Proof: Suppose the given condition holds for two metrics d_1 and d_2. We want to show that an open set on one of the metric spaces, say $U \subset (X, d_1)$, is also open on the other metric space. We know that since U is open in (X, d_1), there exists an open ball $B_\epsilon(u) \subset U$ defined by d_1 for some point $u \in U$. This means that for every point $p \in B_\epsilon(u)$, $d_1(p, u) < \epsilon$. From the condition, we know that for a given $\epsilon > 0$, there must exist a $\delta > 0$ such that $d_2(p, u) < \delta \implies d_1(p, u) < \epsilon$, i.e. that the open ball $B_\delta(u)$ defined by d_2 is contained in U. This thus holds for every point $u \in U$. Therefore, U is open in (X, d_2).

Problems

Problem 1

Consider the set $C[0, 1]$ of continuous real functions on $[0, 1]$. Show that

$$d(f, g) = \int_0^1 |f(x) - g(x)|dx$$

defines a metric on $C[0, 1]$. Does this still hold if continuity is weakened to integrability .

We clearly see that $|f(x) - g(x)| \geq 0$ for $f, g \in C[0, 1]$ and $x \in [0, 1]$ and so

$$d(f, g) \geq 0$$

Similarly, $|f(x) - g(x)| = |g(x) - f(x)|$ and so

$$d(f, g) = d(g, f)$$

Now, consider $f, g, h \in C[0, 1]$. Recall that for a point $x \in [0, 1]$, $|f(x) - g(x)| \leq |f(x) - h(x)| + |h(x) - g(x)|$ and so

$$d(f, g) \leq d(f, h) + d(h, g)$$

Thus, d defines a metric on $C[0, 1]$.

Now consider the set of Riemann integrable functions, $R[0, 1]$. Let $f, g, h \in R[0, 1]$. We know that the Lebesgue's criterion states that a function f is Riemann integrable if and only if the measure of the set of its discontinuities $\mu(D_f)$ is zero.

Consider $m = f - g$. We know clearly that D_f and D_g have measure zero, and we also know that $D_m \subset D_f \cup D_g$ meaning $\mu(D_m) = 0$. Therefore, m is Riemann integrable and

$$d(f,g) = \int_0^1 |f(x) - g(x)|dx = \int_0^1 |m(x)|dx \geq 0$$

It is self evident that for $f, g \in R[0,1]$, $d(f,g) = d(g,f)$. Finally, via similar construction as in the first case, we know that the new functions $m = f - g$, $n = f - h$ and $p = h - g$ are also Riemann integrable and so $d(f,g) \leq d(f,h) + d(h,g)$.

Checking this online, it turns out the definition of the metric space given actually defines a pseudometric space. This is because two distinct functions f, g can still result in $d(f,g) = 0$

Problem 2

If X is a metric space and $x_0 \in X$, show that $f : X \to \mathbb{R}$ given by $f(x) = d(x, x_0)$ is continuous.

From the triangle inequality, we know that for the points $x, y, x_0 \in \mathbb{R}$, $d(x,x_0) \leq d(x,y) + d(y,x_0)$ and $d(y,x_0) \leq d(y,x) + d(x,x_0)$ and so $d(x,y) \geq |d(y,x_0) - d(x,x_0)|$, i.e. $|f(x) - f(y)| < d(x,y)$. Thus, f is Lipschitz continuous.

Problem 3

If A is a subset of a metric space X, define a function $f : X \to \mathbb{R}$ given by $f(x) = d(x,A) = \inf_{y \in A} d(x,y)$. Show that f is continuous.

From the definition, we have that

$$|f(x) - f(y)| = |d(x,A) - d(y,A)|$$

where

$$d(x,A) = \inf_{a_x \in A} d(x, a_x)$$
$$d(y,A) = \inf_{a_y \in A} d(y, a_y)$$

Notice that $d(x,A) \leq d(x,a)$ for some arbitrary point $a \in A$, and likewise that $d(y,A) \leq d(y,a)$ for some $a \in A$, and so

$$|d(x,A) - d(y,A)| \leq |d(x,a) - d(y,a)|$$

for some arbitrary $a \in A$

The triangle inequality states that $d(m,p) \leq d(m,n) + d(n,p)$, for some arbitrary points $m, n, p \in X$

$$d(m,p) \leq d(m,n) + d(n,p) \implies d(m,p) - d(m,n) \leq d(n,p)$$
$$d(m,n) \leq d(m,p) + d(p,n) \implies d(m,n) - d(m,p) \leq d(p,n)$$

Therefore, $|d(m,n) - d(m,p)| \leq d(n,p)$ for arbitrary points $m, n, p \in X$. With that in mind, we have that $|d(x,a) - d(y,a)| \leq d(x,y)$. Therefore, $|f(x) - f(y)| \leq d(x,y)$, and so f is Lipschitz continuous.

2 Topological Spaces

Sometimes, we would like to study spaces spaces without regard to metric, and since the things to study involve open sets, we introduce the concept of the topological space.

2.1 Definition of Topological Space

A topological space is a set X together with a collection of subsets of X called open sets such that

1. the intersection of two open sets is open
2. the union of any collection of open sets is open
3. the empty set $\emptyset$ and the whole space X is open

A set C is called closed if $X - C$ is open.

2.2 Definition of Continuity on Topological Spaces

If X and Y are topological spaces and $f : X \to Y$ is a function, then f is said to be continuous if $f^{-1}(U)$ is open for every open set $U \subset Y$.

It follows from this definition and the definition of closed sets that f is continuous if and only if $f^{-1}(C)$ is closed for every closed set $C \subset Y$.
Proof: Suppose that $f : X \to Y$ is continuous. Clearly then $f^{-1}(U)$ is open for every open set $U \subset Y$. By definition of closed sets, we can generate every closed set by $C = Y - U$. The final step is to show that $f^{-1}(C)$ is closed. To show this, we first must show that $f^{-1}(Y - U) = X - f^{-1}(U)$. To do this, suppose to the contrary that this was not the case, i.e. either there exists an element x both in $f^{-1}(U)$ and $f^{-1}(Y - U)$, or there exists an element not in either one of them. Consider the first scenario. If $x \in f^{-1}(U)$ and $x \in f^{-1}(Y - U)$, then $f(x) \in U$ and $f(x) \in Y - U$, a contradiction. Now consider the second scenario. If $x \notin f^{-1}(U)$ and $x \notin f^{-1}(Y - U)$, then $f(x) \notin U$ and $f(x) \notin Y - U$, which is also a contradiction. Thus, $f^{-1}(Y - U) = X - f^{-1}(U)$, and since $f^{-1}(U)$ is open, $f^{-1}(Y - U) = f^{-1}(C)$ is closed. This completes the proof.

2.3 Definition of Neighborhoods

If X is a topological space and $x \in X$, then a set N is called a neighborhood of x in X if there exists an open set $U \subset N$ such that $x \in N$.

2.4 Definition of Neighborhood Basis

If X is a topological space and $x \in X$, then a collection B_x of subsets of X containing x is called a neighborhood basis at x in X if each neighborhood of x in X contains some element of B_x and each element of B_x is a neighborhood of x.